

KIOSK

Control safety messages simply, quickly and flexibly

The Kiosk system provides a straightforward solution for displaying warning and safety messages on public screens.

The problem:

The challenge at temporary events

At large temporary events - such as festivals, open-air events, sporting events or trade fairs at changing locations - safety managers repeatedly face the same challenge:

How can safety messages be communicated to the audience quickly and reliably in an emergency without permanent infrastructure?

While permanent venues often have suitable systems installed, such solutions are usually unavailable at temporary events. The result is often an improvised setup - for example, a locally connected PC running PowerPoint that is difficult to access and cannot be controlled centrally.

The solution:

The network-based Kiosk system enables simple communication in exceptional situations - reliably and ready for immediate use. Your benefits at a glance:

- **Central control**
Safety messages can be triggered from any device with a Firefox web browser on the same network - from the safety control centre or from a mobile device on site.
- **Password-protected**
The control interface is secured and accessible only to authorised users.
- **Minimal hardware, maximum impact**
A small Windows PC is all that is required. Connect it via HDMI to the video control system for the large screens.
- **Ready for immediate use**
Once prepared images or videos have been stored and configured, the system can be used without additional software. The system displays finished content only; it does not edit or create media.
- **Independent of the internet**
The system operates entirely within the local network; no internet connection is required.
- **Local control when required**
The system can also be operated locally using a Stream Deck, even without a network connection.

Conclusion:

Maximum safety with minimum effort. The Kiosk system enables you to respond quickly and reliably in an emergency.

Technical details - Kiosk system with Windows PC - 1.4.4

The Kiosk system is designed to provide high reliability in safety-critical applications with minimal technical effort.

Hardware, connections and power supply

- **Software requirements:**

Windows 10 or 11, Python 3.12.6 and Firefox.

Important: The system works only with Firefox and Python <= 3.12.6. Chrome no longer permits automatic video playback by default; Firefox can be configured accordingly. The Stream Deck library used by the system does not yet support the latest Python versions. The hardware is supplied with Firefox and Python 3.12.6.

- **Computer**

The system runs on a micro PC with Windows 10 or 11.

- **Network**

The network connection is made directly via a LAN cable.

- **Power**

The PC is powered by an external power adapter.

- **IP address**

After startup, the PC obtains a network address via DHCP. This address is required to control the system. By default, it appears in the lower-right corner of the presentation screen and is hidden after one minute. The second network port uses a static network address.

Media output

- **HDMI output**

The HDMI output starts automatically in kiosk mode.

- **Video encoding**

Videos should be encoded in H.264 / .mp4 format at <= 10 Mbit/s.

Reliability in the event of a fault

- **Automatic restart**

If the network connection is interrupted or the system restarts, the application restarts automatically.

- **Default message after restart**

The system always returns to the preset default message, not the most recently displayed message.

Easy operation and maintenance on the device

Content can be managed or replaced using File Explorer. To change content on site:

- **Close the application**

Press ALT + F4.

- **Update content**

Add, replace or delete videos and images manually.

- **Restart**

Click Kiosk.bat on the desktop.

- **Automatic reconnection**

External Remote applications reconnect automatically after a restart.

Stream Deck

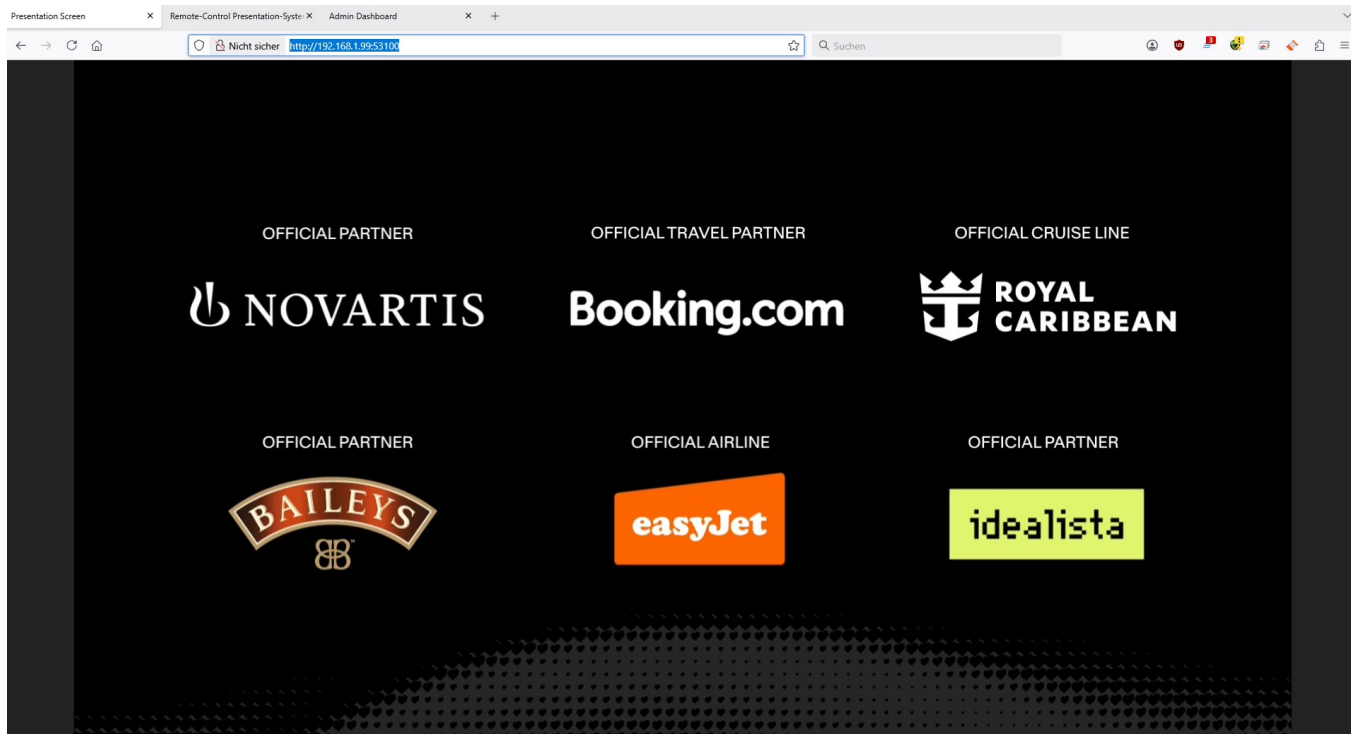
- **Automatic detection**
When a Stream Deck is connected to the PC, the system detects and integrates it automatically.
- **Automatic key assignment**
The keys are assigned automatically. The Stream Deck software does not need to be installed. If installed, make sure it does not start with Windows and is not running while the Kiosk system is operating. Any Stream Deck model (Mini, Midi or Maxi) can be used.
- **Control setting**
By default, the Stream Deck only indicates which clip is selected. To control clips from the Stream Deck, enable Stream Deck control in the Admin Panel. This setting is NOT saved.

Images



Presentation Screen

The Presentation Screen is available on the network at: `http://<IP-address>:53100`
Replace <IP-address> with the address displayed by the PC (e.g. 192.168.1.42:53100).

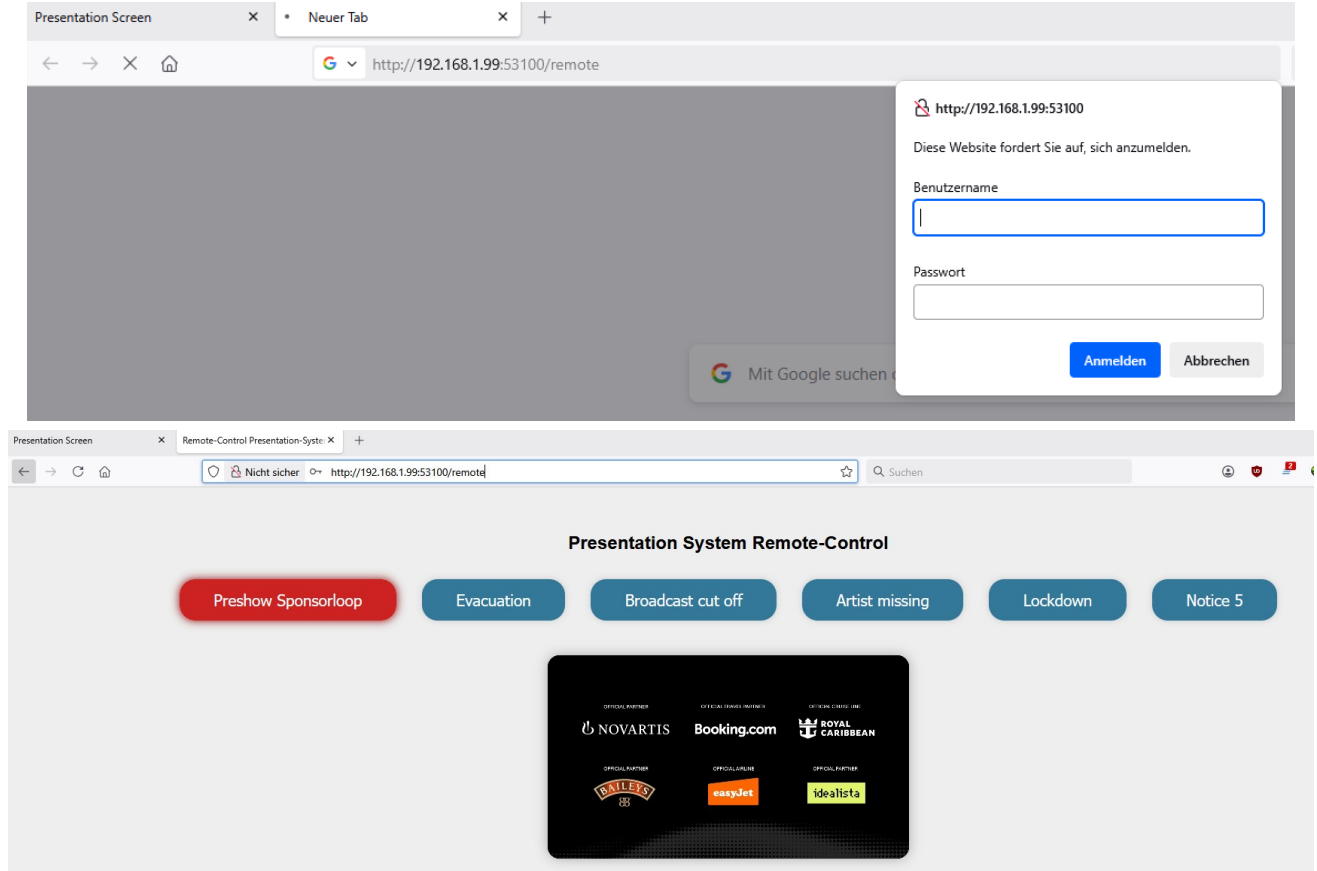


Remote Screen with login

The Remote Screen is available on the network at: `http://<IP-address>:53100/remote`

Replace `<IP-address>` with the address displayed by the PC (e.g. `192.168.1.42:53100/remote`).

The Remote Screen shows a small preview of the currently running programme.



Admin Dashboard

The Admin Dashboard is available on the network at: `http://<IP-address>:53100/admin`

Like the Remote Screen, the Admin Dashboard is password-protected.

Replace `<IP-address>` with the address shown in the desktop terminal window (e.g. `192.168.1.42:53100/admin`).

The Admin Dashboard shows whether the server is running, which item is selected, and allows the buttons and their media to be configured. After saving, the Remote Screen and Stream Deck controls update automatically.

Allow StreamDeck control is not saved and is always disabled at startup.

← → ↻ 🏠 ⚠ Nicht sicher 192.168.1.102:53100/admin

Presentation Screen Admin Dashboard

Current media type: **video**

Media source: **static/videos/video1.mp4**

Presentation client connected: **No**

Remote client connected: **Yes**

Scheduled event (daily at 02:00): **videos/video1.mp4**

Server status: **[OK] Server online (19:03:32)** ●

Reset to default programPlay scheduled medium now

Button-Konfiguration

☒ Aktiv	Preshow Sponsor Loop	Video ▾	videos/video1.mp4
☒ Aktiv	Evacuation	Slideshow ▾	bild1/
☒ Aktiv	Broadcast Cut Off	Slideshow ▾	bild2/
☒ Aktiv	Artist missing	Slideshow ▾	bild3/
☒ Aktiv	Lock down	Slideshow ▾	bild4/
☐ Aktiv	Notice 5	Slideshow ▾	bild5/
☐ Aktiv	Notice 6	Slideshow ▾	bild6/
☐ Aktiv	Notice 7	Slideshow ▾	bild7/
☐ Aktiv	Notice 8	Slideshow ▾	bild8/
☐ Aktiv	Notice 9	Slideshow ▾	bild9/
☐ Aktiv	Notice 10	Slideshow ▾	bild10/
☒ Aktiv	Video2	Video ▾	videos/video2.mp4
☐ Aktiv	Video3	Video ▾	videos/video3.mp4
☐ Aktiv	Video4	Video ▾	videos/video4.mp4
☐ Aktiv	Video5	Video ▾	videos/video5.mp4
☐ Aktiv	Playlist	Playlist ▾	

Save Button Configuration☐ Allow StreamDeck control

Admin Dashboard - Button configuration guide

The Admin Panel provides central control and configuration of the presentation system's media buttons. It defines which content is displayed and controlled on the connected Stream Deck.

Settings available for each button

- **Activation**
A checkbox determines whether the button is visible and functional on the Stream Deck.
- **Label**
The text shown on the button. Multi-line text is wrapped automatically at spaces.
- **Type**
Select Slideshow (image folder), Video (single video file), or Playlist (sequence of media items).
- **Source (src)**
Path to the media file or folder, for example bild5 or videos/video1.mp4.
- **Audio**
Videos may contain audio. Audio is output through the playback computer using the Windows default audio device.

Operation

- **Load configuration**
The current button configuration is loaded automatically when the Admin Panel is opened.
- **Edit**
Activation, label, type and source can be changed directly in the interface.
- **Save**
Click the Save button to apply changes.

Behaviour after saving

- **Configuration file**
button_config.json is updated.
- **Real-time update**
All connected devices receive the updated configuration immediately.
- **Stream Deck display**
Active buttons are reloaded. Disabled buttons appear blank (black) and have no function.

Configuration notes

- **Layout**
Inactive buttons do not appear in the active layout.
- **Order**
The button order follows the configuration file. Inactive buttons are skipped so active buttons are displayed without gaps.
- **No restart required**
Changes made in the Admin Panel take effect immediately.

Local control using a Stream Deck

The application can also be operated directly on the device using a Stream Deck. The connected model is detected automatically. A Stream Deck can also be connected later; the keys appear after approximately 10 seconds.

The Stream Deck key assignments are synchronised with the Remote application. This means:

- **Remote to Stream Deck**
The Stream Deck follows commands from the Remote interface.
- **Stream Deck to Remote**
The Remote interface receives commands triggered locally on the Stream Deck.
- **Control setting**
By default, the Stream Deck only indicates which clip is selected. To control clips, enable Stream Deck control in the Admin Panel. This setting is NOT saved.

This keeps operation flexible - either centrally over the network or directly on site, independently of network access.



Content and media formats

The system supports two types of content: images and videos.

Image content

- **Supported formats**
.jpg, .jpeg, .png, .gif and .bmp
- **Directory**
kiosk/static/<foldername> (e.g. bild1, bild2, ...)
- **Updating images**
Desktop shortcuts are provided. Copy new images into the relevant folders and restart the PC.
- **Filenames**
Image filenames can be chosen freely.
- **Slideshow**
Multiple images are displayed in an endless loop at five-second intervals. Configure the message and associated control button in the Admin application.

Video content for individual videos

- **Supported format**
.mp4
- **Directory**
kiosk/static/videos/
- **Naming scheme**
video1.mp4, video2.mp4, video3.mp4, etc. Custom filenames are not supported.

Video content for the video playlist

- **Supported format**
.mp4
- **Directory**
kiosk/static/videos_playlist/
- **Playback**
Copy the videos into this folder. They are played one after another in a loop, in alphabetical order. Only one video playlist is available.

API control via HTTP

The system can be controlled directly through a simple HTTP interface. Any application capable of sending HTTP requests can display configured content. This is suitable for integration into existing safety systems, automation tools or custom control solutions.

Example calls:

- **http://<IP*>:53100/api?Function=bild1**
Starts the slideshow using the images in the bild1/ folder.
- **http://<IP*>:53100/api?Function=video1**
Plays the predefined sponsor video.
- **http://<IP*>:53100/api?Function=reset**
Restores the preset default media.
- **http://<IP*>:53100/api?Function=playlist**
Starts the configured playlist.

*Replace <IP> with the current IP address of the PC on the network.

The relevant content must first be stored and configured in the system. The API is available while the Kiosk application is running.

Computer login

PC

No password is set. Simply click Sign in.

Kiosk app

User: admin

Password: password123

The password for the Kiosk app can be changed in app.py using a text editor. Be careful not to make any other changes to the code.